



# Biological Responses in Coastal and Marine Systems

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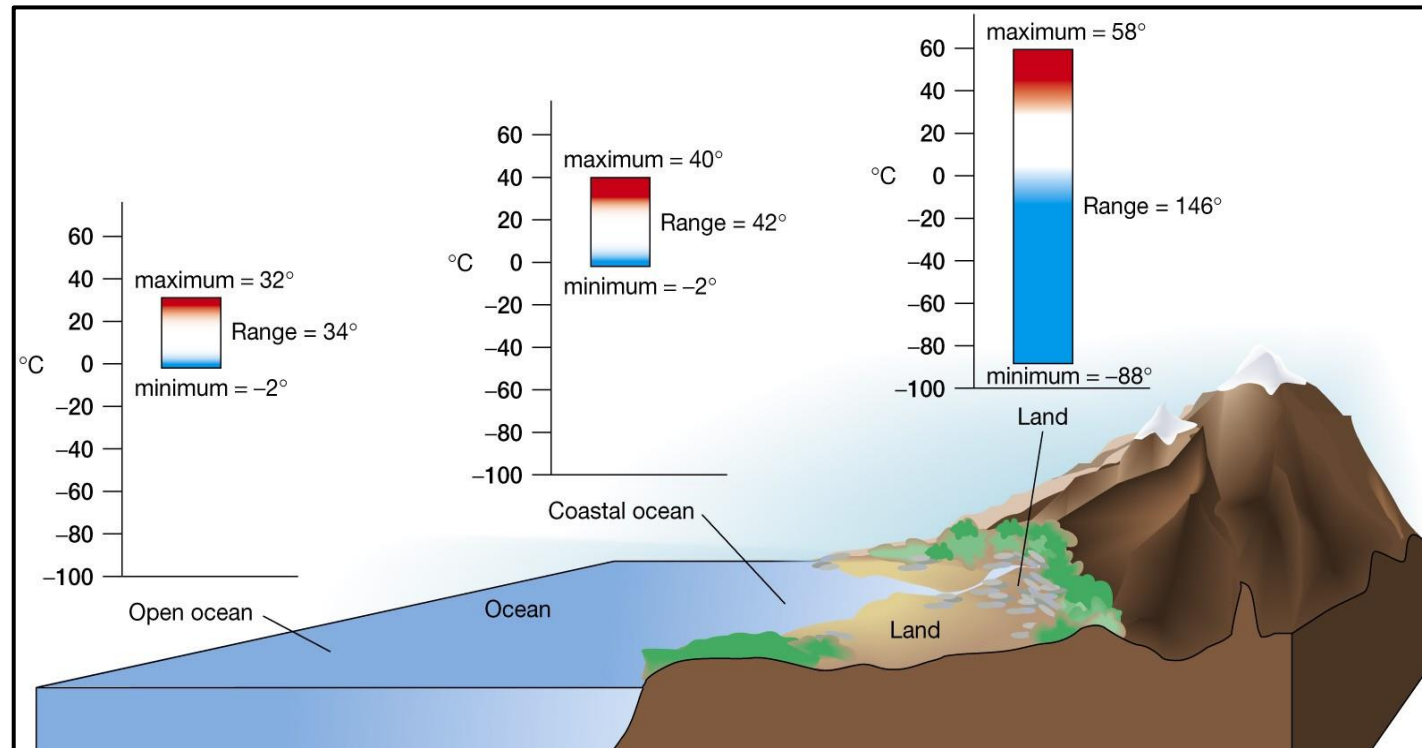
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# MARINE VS TERRESTRIAL SYSTEM



- Conditions in the marine environment are relatively uniform compared to land
- Little seasonal variation in marine environments
- Less oxygen (~ 30%) available in seawater
- Marine ectotherms are more susceptible to warming
  - Have narrower thermal safety margins than terrestrial ectotherms

# MARINE VS TERRESTRIAL SYSTEM



- Thermal tolerance breadth narrower
- Live closer to thermal limits, making species more sensitive to warming
- Few thermal escapes (i.e. nowhere to go, as temp are more uniform)

- Thermal tolerance breadth wider
- Greater habitat heterogeneity allows for more behavioral/physiological adaptation options

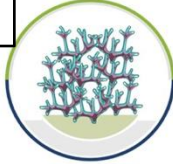
# **Responses to Ocean Warming**



| Marine & Coastal Ecosystems                                                                                                                                                                                       | RESPONSE VARIABLE CATEGORIES | CLIMATE CHANGE       |                        |                                           |                     |                |                  |         |                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------|------------------------|-------------------------------------------|---------------------|----------------|------------------|---------|----------------------------|
|                                                                                                                                                                                                                   |                              | Warming / heat waves | Extreme weather events | Atmospheric CO <sub>2</sub> concentration | Ocean acidification | Sea level rise | Salinity (hyper) | Cooling | Rainfall regimes (reduced) |
| <div>Caribbean</div> <div>  <div>CORAL REEFS</div> </div>                                                                         | Density/Abundance            | 20%                  | 25%                    |                                           | 4%                  |                |                  |         |                            |
|                                                                                                                                                                                                                   | Distribution                 |                      |                        |                                           |                     |                |                  |         |                            |
|                                                                                                                                                                                                                   | Growth                       | 10%                  | 8%                     |                                           | 32%                 | 20%            |                  |         |                            |
|                                                                                                                                                                                                                   | Habitat structure            | 21%                  | 42%                    |                                           | 18%                 | 80%            |                  |         |                            |
|                                                                                                                                                                                                                   | Metabolism                   | 6%                   |                        |                                           | 7%                  |                |                  |         |                            |
|                                                                                                                                                                                                                   | Photosynthesis/Respiration   | 9%                   |                        |                                           | 18%                 |                |                  |         |                            |
|                                                                                                                                                                                                                   | Reproduction                 | 14%                  | 8%                     |                                           | 14%                 |                |                  |         |                            |
|                                                                                                                                                                                                                   | Other                        | 19%                  | 17%                    |                                           | 7%                  |                |                  |         |                            |
|                                                                                                                                                                                                                   | Total number of studies      | 78                   | 12                     |                                           | 28                  | 5              |                  |         |                            |
|                                                                                                                                                                                                                   |                              |                      |                        |                                           |                     |                |                  |         |                            |
| <div>MANGROVE FORESTS</div> <div>  <div>MANGROVE FORESTS</div> </div>                                                             | Density/Abundance            | 7%                   | 17%                    |                                           |                     | 8%             | 16%              | 63%     | 21%                        |
|                                                                                                                                                                                                                   | Distribution                 | 10%                  |                        |                                           |                     | 23%            |                  |         | 7%                         |
|                                                                                                                                                                                                                   | Growth                       | 46%                  | 22%                    | 100%                                      |                     | 54%            | 50%              | 5%      | 36%                        |
|                                                                                                                                                                                                                   | Habitat structure            | 2%                   | 22%                    |                                           |                     |                |                  | 16%     | 36%                        |
|                                                                                                                                                                                                                   | Metabolism                   | 5%                   |                        |                                           |                     | 15%            | 8%               |         |                            |
|                                                                                                                                                                                                                   | Photosynthesis/Respiration   | 29%                  | 28%                    |                                           |                     |                | 26%              | 16%     | 33%                        |
|                                                                                                                                                                                                                   | Reproduction                 |                      | 11%                    |                                           |                     |                |                  |         |                            |
|                                                                                                                                                                                                                   | Other                        |                      |                        |                                           |                     |                |                  |         |                            |
|                                                                                                                                                                                                                   | Total number of studies      | 42                   | 18                     | 15                                        |                     | 13             | 39               | 19      | 14                         |
|                                                                                                                                                                                                                   |                              |                      |                        |                                           |                     |                |                  |         | 9                          |
| <div>SEAGRASS BEDS<br/><i>Thalassia testudinum</i></div> <div>  <div>SEAGRASS BEDS<br/><i>Thalassia testudinum</i></div> </div> | Density/Abundance            | 20%                  | 33%                    |                                           |                     |                | 20%              |         |                            |
|                                                                                                                                                                                                                   | Distribution                 | 20%                  | 67%                    |                                           |                     | 100%           |                  |         |                            |
|                                                                                                                                                                                                                   | Growth                       | 20%                  |                        |                                           |                     |                | 20%              |         |                            |
|                                                                                                                                                                                                                   | Habitat structure            |                      |                        |                                           |                     |                |                  |         |                            |
|                                                                                                                                                                                                                   | Metabolism                   | 20%                  |                        |                                           |                     |                | 20%              |         |                            |
|                                                                                                                                                                                                                   | Photosynthesis/Respiration   | 20%                  |                        |                                           |                     |                | 40%              |         |                            |
|                                                                                                                                                                                                                   | Reproduction                 |                      |                        |                                           |                     |                |                  |         |                            |
|                                                                                                                                                                                                                   | Other                        |                      |                        |                                           |                     |                |                  |         |                            |
|                                                                                                                                                                                                                   | Total number of studies      | 5                    | 3                      |                                           |                     | 1              | 10               |         |                            |
|                                                                                                                                                                                                                   |                              |                      |                        |                                           |                     |                |                  |         |                            |

Effects of climate change on marine coastal ecosystems – A review to guide research and management

Ewan Trégarot <sup>a,\*</sup>, Juan Pablo D'Oliveiro <sup>b</sup>, Andrea Zita Botelho <sup>c,d</sup>, Andrea Cabrito <sup>c</sup>, Gabriel O. Cardoso <sup>b</sup>, Gema Casal <sup>e</sup>, Cindy C. Cornet <sup>b</sup>, Simon M. Cragg <sup>f</sup>, A. Karima Degia <sup>b</sup>, Stein Fredriksen <sup>g</sup>, Elisa Furlan <sup>h,k</sup>, Georg Heiss <sup>b</sup>, Diego K. Kersting <sup>l</sup>, Jean-Philippe Maréchal <sup>m</sup>, Erik Meesters <sup>n</sup>, Bethan C. O'Leary <sup>o,p</sup>, Géraldine Pérez <sup>q</sup>, Cristina Seijo-Núñez <sup>r</sup>, Remy Simide <sup>q</sup>, Matthijs van der Geest <sup>n</sup>, Silvia de Juan <sup>s</sup>

| Marine & Coastal Ecosystems                                                                                                                                                                              | RESPONSE VARIABLE CATEGORIES | CLIMATE CHANGE       |                        |                                           |                     |                |                 |         |                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------|------------------------|-------------------------------------------|---------------------|----------------|-----------------|---------|----------------------------|
|                                                                                                                                                                                                          |                              | Warming / heat waves | Extreme weather events | Atmospheric CO <sub>2</sub> concentration | Ocean acidification | Sea level rise | Salinity (hypo) | Cooling | Rainfall regimes (reduced) |
| <div>North-East Atlantic</div> <div>  <div>KELP FORESTS</div> </div>                                                  | Density/Abundance            | 9%                   |                        |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Distribution                 | 23%                  |                        |                                           |                     |                | 58%             |         |                            |
|                                                                                                                                                                                                          | Growth                       | 20%                  |                        |                                           |                     |                | 8%              |         |                            |
|                                                                                                                                                                                                          | Habitat structure            | 2%                   | 100%                   |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Metabolism                   | 16%                  |                        |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Photosynthesis/Respiration   | 17%                  |                        |                                           |                     |                | 17%             |         |                            |
|                                                                                                                                                                                                          | Reproduction                 | 13%                  |                        |                                           |                     |                | 17%             |         |                            |
|                                                                                                                                                                                                          | Other                        |                      |                        |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Total number of studies      | 179                  | 1                      |                                           |                     |                | 12              |         |                            |
|                                                                                                                                                                                                          |                              |                      |                        |                                           |                     |                |                 |         |                            |
| <div>MAËRL BEDS</div> <div>  <div>MAËRL BEDS</div> </div>                                                             | Density/Abundance            |                      |                        |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Distribution                 | 33%                  |                        |                                           | 9%                  |                |                 |         |                            |
|                                                                                                                                                                                                          | Growth                       | 33%                  |                        |                                           | 36%                 |                |                 |         |                            |
|                                                                                                                                                                                                          | Habitat structure            |                      |                        |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Metabolism                   | 33%                  |                        |                                           | 27%                 |                |                 |         |                            |
|                                                                                                                                                                                                          | Photosynthesis/Respiration   |                      |                        |                                           | 27%                 |                |                 |         |                            |
|                                                                                                                                                                                                          | Reproduction                 |                      |                        |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Other                        |                      |                        |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Total number of studies      | 4                    |                        |                                           | 13                  |                |                 |         |                            |
|                                                                                                                                                                                                          |                              |                      |                        |                                           |                     |                |                 |         |                            |
| <div>SALT MARSHES</div> <div>  <div>SALT MARSHES</div> </div>                                                         | Density/Abundance            | 3%                   |                        | 7%                                        |                     | 7%             |                 |         | 6%                         |
|                                                                                                                                                                                                          | Distribution                 |                      |                        |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Growth                       | 16%                  |                        | 21%                                       |                     | 12%            |                 |         |                            |
|                                                                                                                                                                                                          | Habitat structure            | 32%                  |                        | 14%                                       |                     | 52%            |                 |         | 69%                        |
|                                                                                                                                                                                                          | Metabolism                   |                      |                        |                                           |                     | 5%             |                 |         | 19%                        |
|                                                                                                                                                                                                          | Photosynthesis/Respiration   | 41%                  |                        | 29%                                       |                     | 15%            |                 |         | 6%                         |
|                                                                                                                                                                                                          | Reproduction                 | 8%                   |                        | 21%                                       |                     | 8%             |                 |         |                            |
|                                                                                                                                                                                                          | Other                        |                      | 100%                   | 7%                                        |                     | 2%             |                 |         |                            |
|                                                                                                                                                                                                          | Total number of studies      | 37                   | 1                      | 14                                        |                     | 60             |                 |         | 16                         |
|                                                                                                                                                                                                          |                              |                      |                        |                                           |                     |                |                 |         |                            |
| <div>SEAGRASS BEDS<br/><i>Zostera marina</i></div> <div>  <div>SEAGRASS BEDS<br/><i>Zostera marina</i></div> </div> | Density/Abundance            | 18%                  |                        |                                           |                     |                | 15%             |         |                            |
|                                                                                                                                                                                                          | Distribution                 | 3%                   | 100%                   |                                           |                     |                | 8%              |         |                            |
|                                                                                                                                                                                                          | Growth                       | 15%                  |                        |                                           |                     |                | 15%             |         |                            |
|                                                                                                                                                                                                          | Habitat structure            | 9%                   |                        |                                           |                     |                | 15%             |         |                            |
|                                                                                                                                                                                                          | Metabolism                   | 42%                  |                        |                                           |                     |                | 38%             |         |                            |
|                                                                                                                                                                                                          | Photosynthesis/Respiration   | 12%                  |                        |                                           |                     |                | 8%              |         |                            |
|                                                                                                                                                                                                          | Reproduction                 |                      |                        |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Other                        |                      |                        |                                           |                     |                |                 |         |                            |
|                                                                                                                                                                                                          | Total number of studies      | 33                   | 1                      |                                           |                     |                | 13              |         |                            |
|                                                                                                                                                                                                          |                              |                      |                        |                                           |                     |                |                 |         |                            |

Response Legend

Fully Mostly

Negative

Fully Mostly

Positive

Fully Mostly

Neutral

Fully Mostly

Unclear

Mixed

Not tested

Response Legend

Fully Mostly

Negative

Fully Mostly

Positive

Fully Mostly

Neutral

Fully Mostly

Unclear

Mixed

Not tested

# EFFECTS OF **OCEAN WARMING** AT SPECIES LEVEL

- Organisms from stable environments (e.g. tropical or polar regions) are more sensitive to warming
  - because their evolutionary history has led them to thrive in conditions with minimal temperature fluctuations
- In contrast, variable environments (e.g. estuaries and coastal environments) expose organisms to a wider range of temperatures
  - Allows them to develop broader temperature tolerances and acclimatization strategies overtime



**Stable environments**

**vs**



**Variable environments**



# EFFECTS OF OCEAN WARMING AT SPECIES LEVEL

## Biota from **stable** environments

### *Narrow thermal niche*

Have a narrow thermal niche, meaning they are adapted to a specific temperature range and have limited ability to tolerate significant deviations from that range.

### *Limited acclimatization*

May have limited capacity for acclimatization, or physiological adjustments to environmental changes, compared to those from variable environments.

### *Sensitivity to rapid warming*

When faced with rapid warming, they are more likely to experience stress, reduced fitness, and even mortality because their limited thermal tolerances and acclimatization capabilities are overwhelmed.

## Biota from **variable** environments

### *Evolutionary pressure*

Natural selection favors organisms that can tolerate and adapt to a broader range of temperatures, including seasonal fluctuations and occasional extremes.

### *Adaptation mechanisms*

Have evolved diverse adaptation mechanisms, such as behavioral changes (e.g., burrowing, migration), physiological adjustments (e.g., changes in enzyme activity), and genetic adaptations, to cope with fluctuating conditions.

# EFFECTS OF OCEAN WARMING AT SPECIES LEVEL

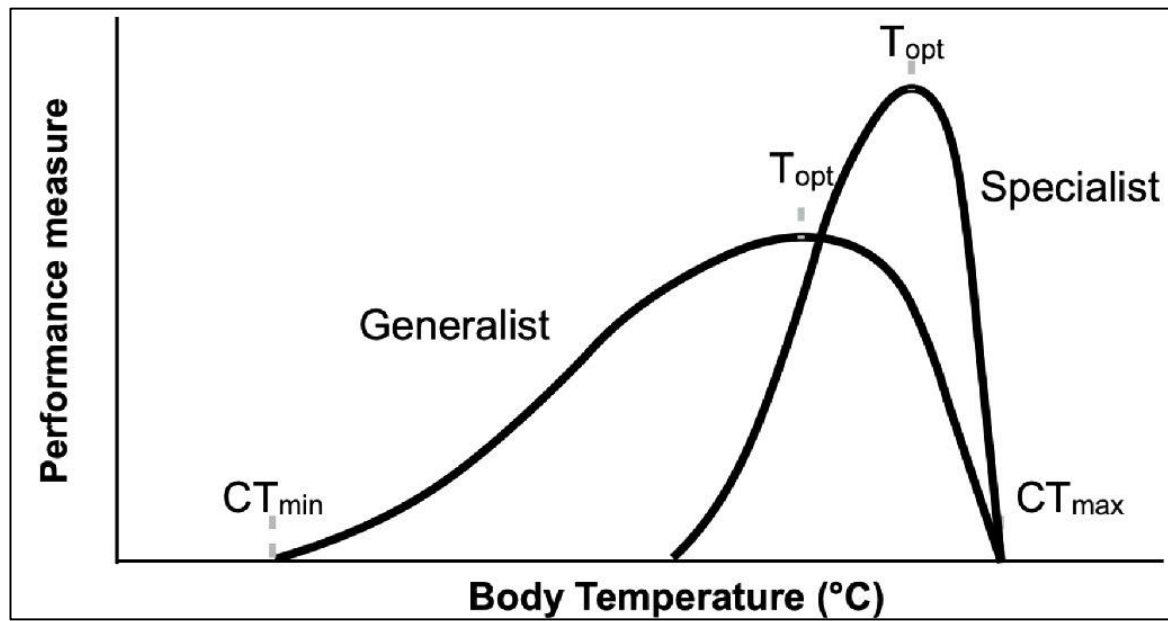


Fig. Hypothetical thermal performance curves

## Specialist organisms

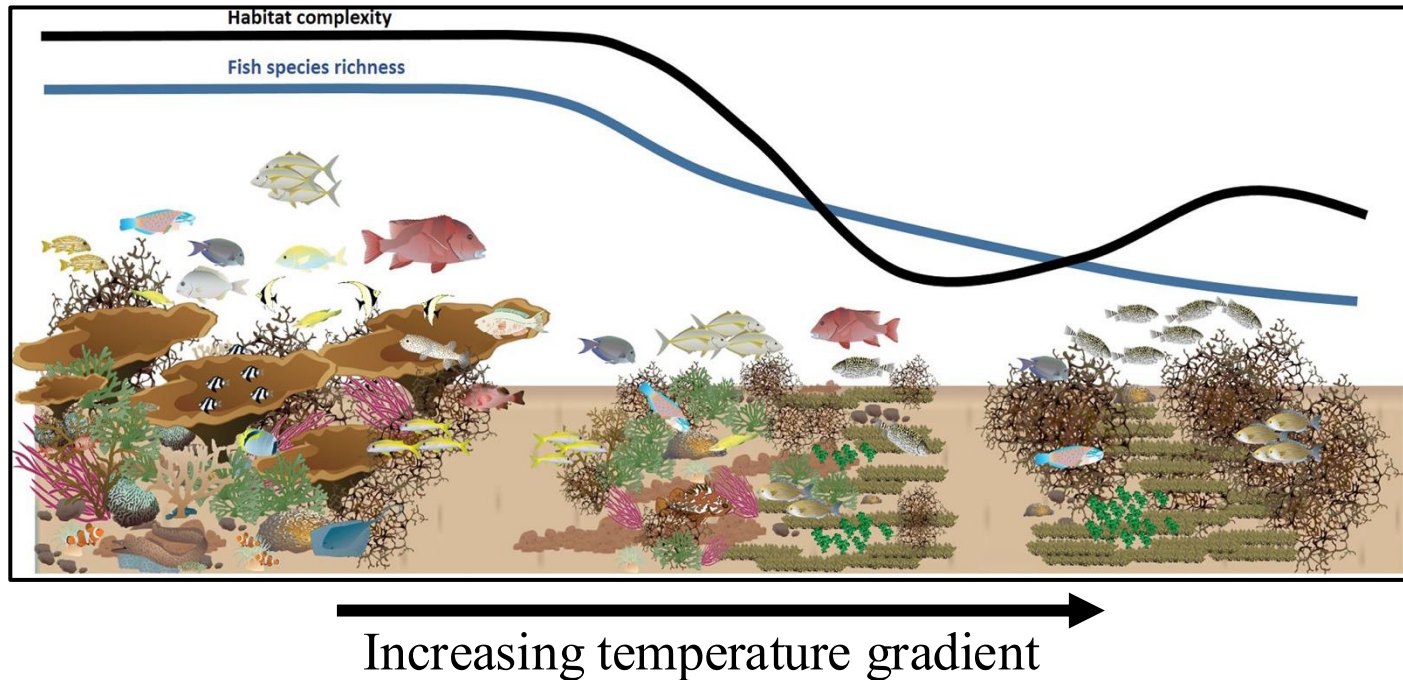
- Narrow temp range
- Organisms withstand small variation in temperature
- Typically live in stable environments (e.g. tropical or polar regions)

## Generalist organisms

- Wide temp range
- Organisms withstand large variation in temperature
- Typically live in variable environments (i.e. estuaries and coastal environments)



# EFFECTS OF OCEAN WARMING AT COMMUNITY LEVEL



*Ecology*, 91(8), 2010, pp. 2198–2204  
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Ocean warming increases threat of invasive species  
in a marine fouling community

CASCADE J. B. SORTE,<sup>1,3</sup> SUSAN L. WILLIAMS,<sup>1</sup> AND ROBYN A. ZEREBECKI<sup>2</sup>

OPEN ACCESS Freely available online

PLoS one

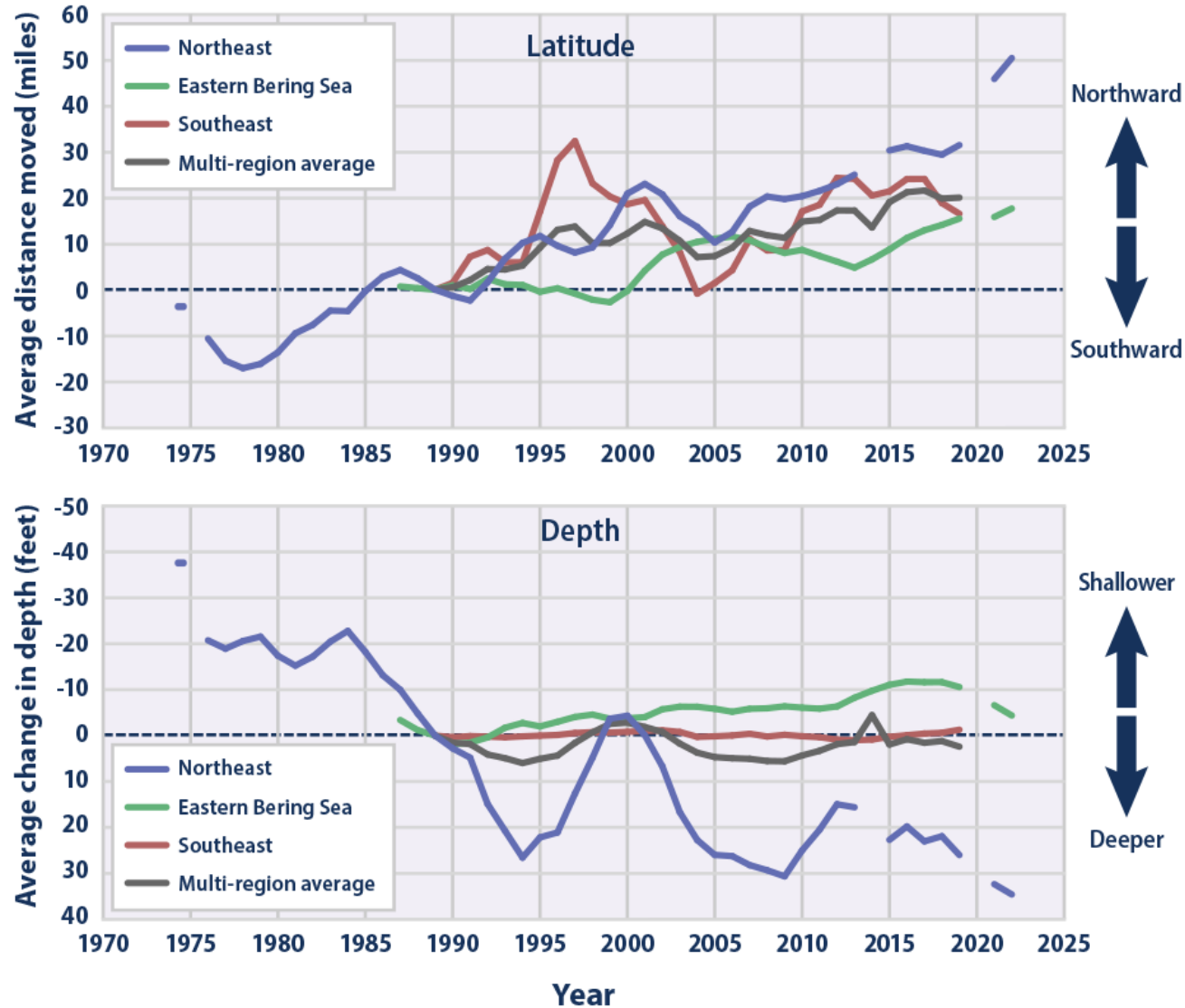
Turning on the Heat: Ecological Response to Simulated  
Warming in the Sea

Dan A. Smale<sup>1\*</sup>, Thomas Wernberg<sup>1,2</sup>, Lloyd S. Peck<sup>3</sup>, David K. A. Barnes<sup>3</sup>

- Alters recruitment (addition of new individuals to the community)
- Reduces species diversity and abundance
- Which leads to changes in overall community structure
- Simplified communities, dominated by species with broad thermal tolerance ranges (i.e. invasive species)
- Which could facilitate competition, where some species are outcompeted by others

# EFFECTS OF OCEAN WARMING ON SPECIES DISTRIBUTION

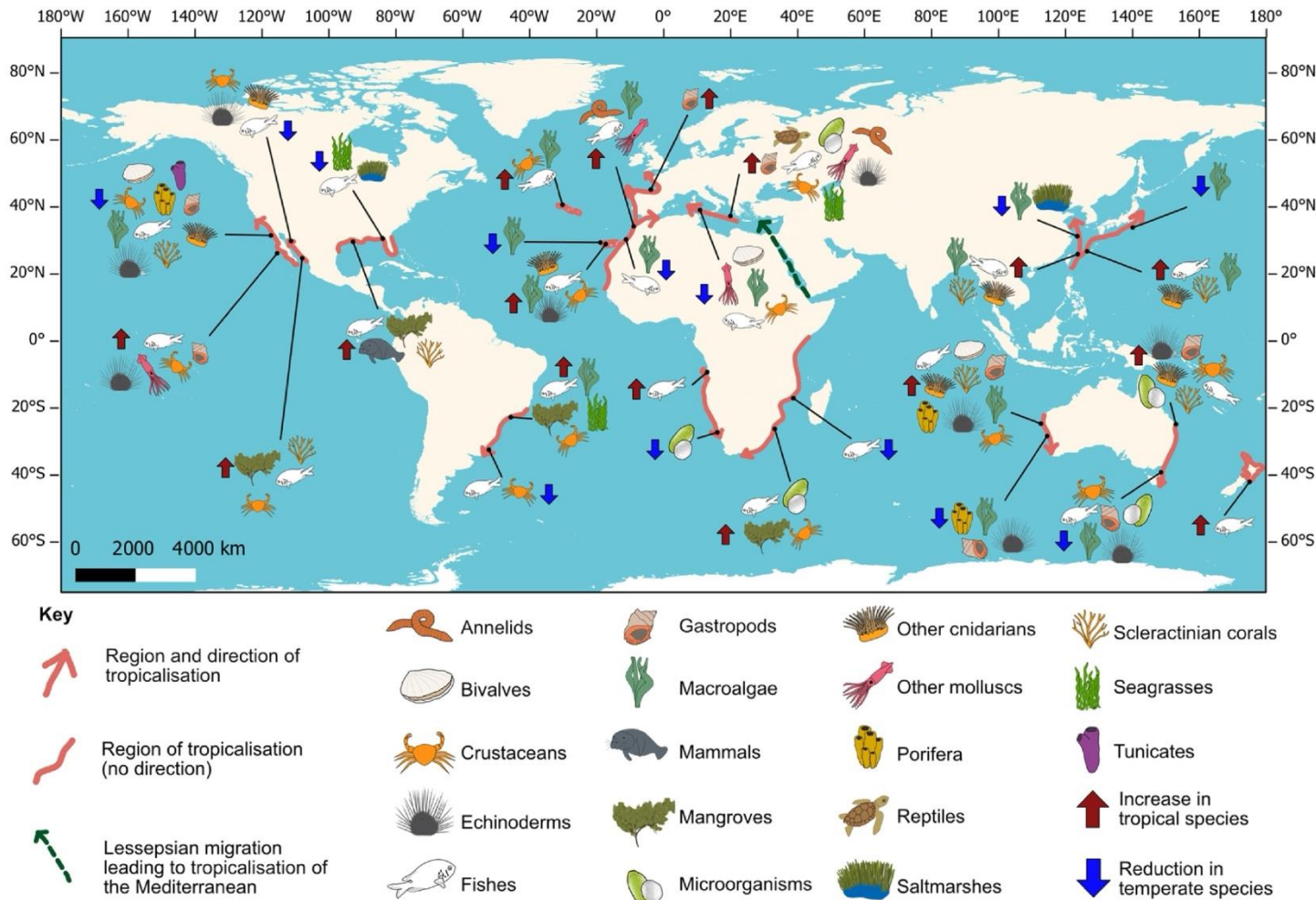
## Poleward and Depthward Shifts



- **Thermal habitat compression:** species are squeezed into narrower ranges
- Species tend to **shift toward the poles** or into deeper, cooler waters to remain within thermal tolerances
- Documented examples:
  - *Atlantic cod* moving northwards in the North Atlantic
  - Tropical fish species now found in subtropical zones

# EFFECTS OF OCEAN WARMING ON SPECIES DISTRIBUTION

## Tropicalisation

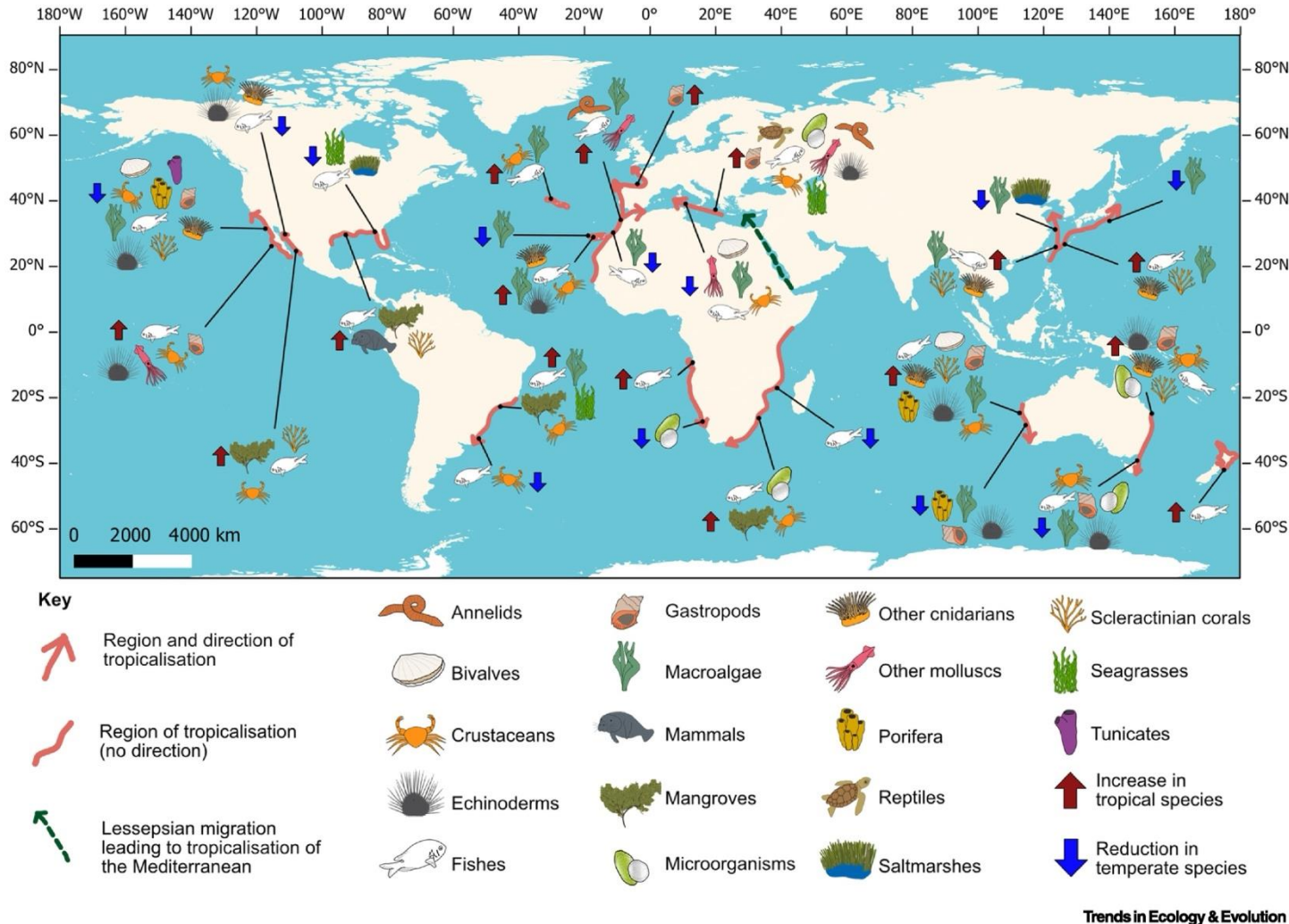


The phenomenon of tropical/subtropical species expanding into temperate regions and temperate species receding/retracting, driven by rising sea temperatures due to climate change.



# EFFECTS OF OCEAN WARMING ON SPECIES DISTRIBUTION

## Tropicalisation



### Driven by:

- *Warm water boundary currents*
  - (which heat faster and distribute species)
- *Marine heatwaves*
  - (which lead to loss of temperate species)
- *Warming winter temperatures*
  - (which usually restrict range)
- Thermal tolerance, dispersal capability, and adaptability of species
- Functional niche availability



# EFFECTS OF **OCEAN WARMING** ON SPECIES DISTRIBUTION

## **Tropicalisation**

### *Ecological Consequences*

#### **1. Loss of temperate ecosystems and species**

- ***Seaweed Forests Decline:*** Temperate seaweed forests, crucial habitats for numerous species, can be replaced by less complex ecosystems or even barren grounds as tropical herbivores and algae proliferate
- ***Retraction of Temperate Species:*** As tropical organisms expand into temperate regions, native temperate species may be outcompeted or unable to adapt, leading to their decline or disappearance from these areas.

# EFFECTS OF **OCEAN WARMING** ON SPECIES DISTRIBUTION

## **Tropicalisation**

### *Ecological Consequences*

## **2. Changes in Species Interactions**

- *Altered Community Composition:* The influx of tropical species can lead to changes in competition, predation, and herbivory, reshaping food webs and altering the overall trophic structure of temperate ecosystems.
- *New Trophic Pathways:* The introduction of new organisms can create novel ecological pathways and interactions, some of which may be beneficial, while others may be detrimental to the existing ecosystem structure.

# EFFECTS OF **OCEAN WARMING** ON SPECIES DISTRIBUTION

## **Tropicalisation**

### *Ecological Consequences*

#### **3. Ecosystem Transformation**

- ***Phase Shifts:*** Tropicalisation can drive entire ecosystems, such as temperate reefs, to undergo "phase shifts" where they transition from one stable state (e.g., kelp-dominated) to another, often characterized by different species and reduced biodiversity.
- ***Modified Ecosystem Functions:*** The changes in species composition and structure affect broader ecosystem functions, like carbon storage and the overall flow of energy and materials within the ecosystem.

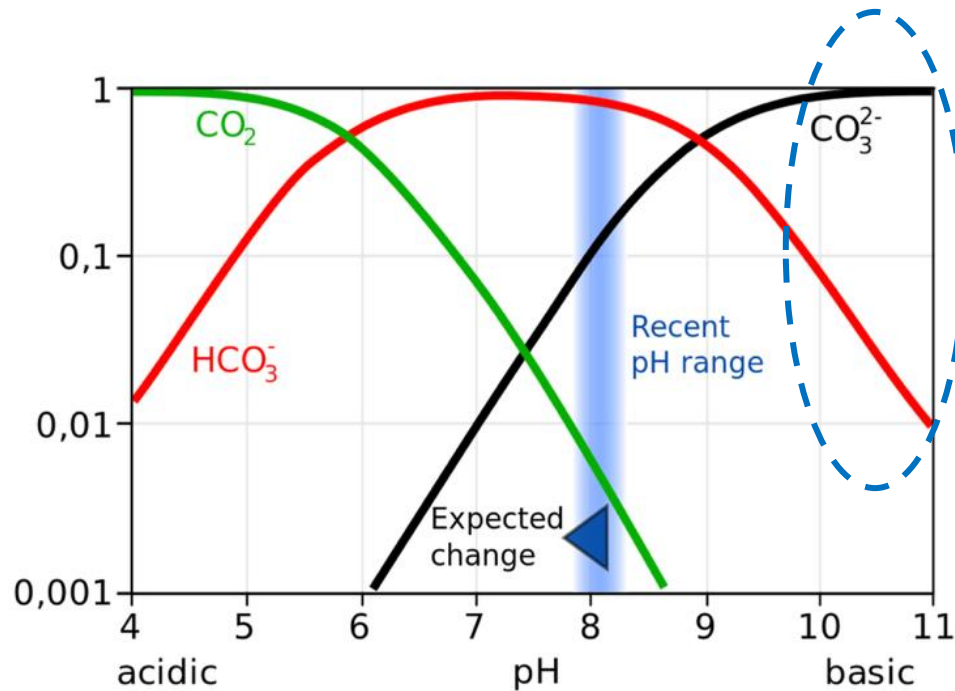




## Responses to Ocean Acidification

# EFFECTS OF OCEAN ACIDIFICATION AT SPECIES LEVEL

## Calcification



- **Marine organisms** such as foraminifera, coccolithophores, corals build shells/skeletons:
  - eqn 1:  $\text{Ca}^{2+} + \text{CO}_3^{2-} \rightarrow \text{CaCO}_3$
- Occurs under **high pH** conditions when there is sufficient calcium ions
- However, the global current pH  $\sim 8.1$
- Ocean pH is not high enough for  $\text{CO}_3^{2-}$  to dominate, so the direct  $\text{Ca}^{2+} + \text{CO}_3^{2-}$  reaction is less relevant under current seawater pH conditions

# EFFECTS OF OCEAN ACIDIFICATION AT SPECIES LEVEL

## Calcification

- At **near – neutral pH**  $\text{HCO}_3^-$  ions are more dominant



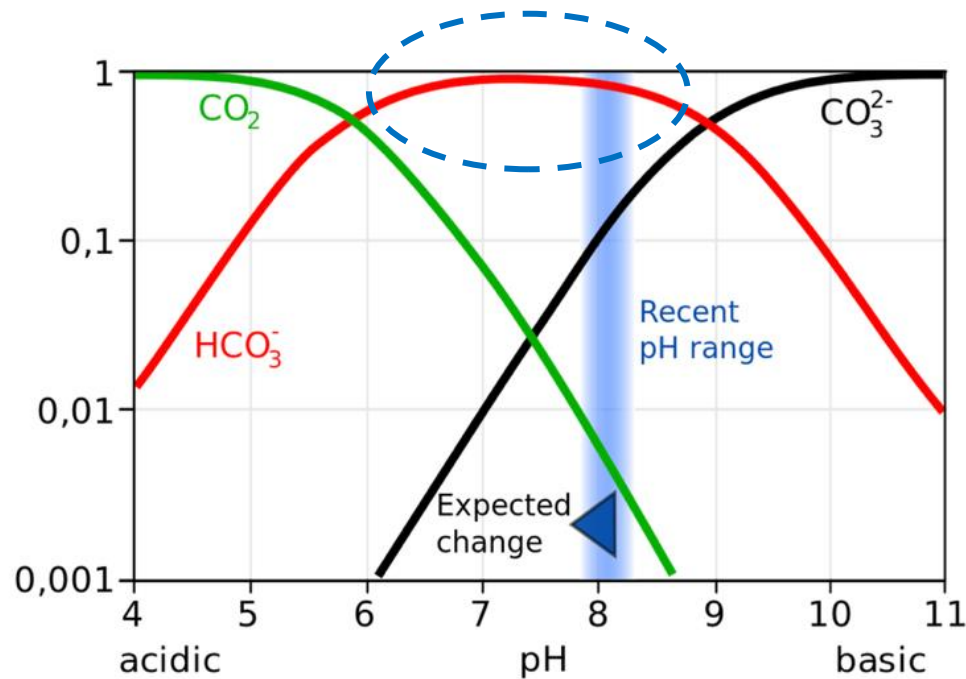
- $\text{H}_2\text{CO}_3$  is weak and unstable
- Therefore, it dissociates to form  $\text{H}_2\text{O}$  and  $\text{CO}_2$ , which increases  $\text{CO}_2$  and acts as a carbonate counter pump:



- Eqns 2 & 3 result in a net reaction:



- $\text{CaCO}_3$  precipitation removes ionic forms of DIC  
→ decrease pH → increase  $\text{CO}_2$

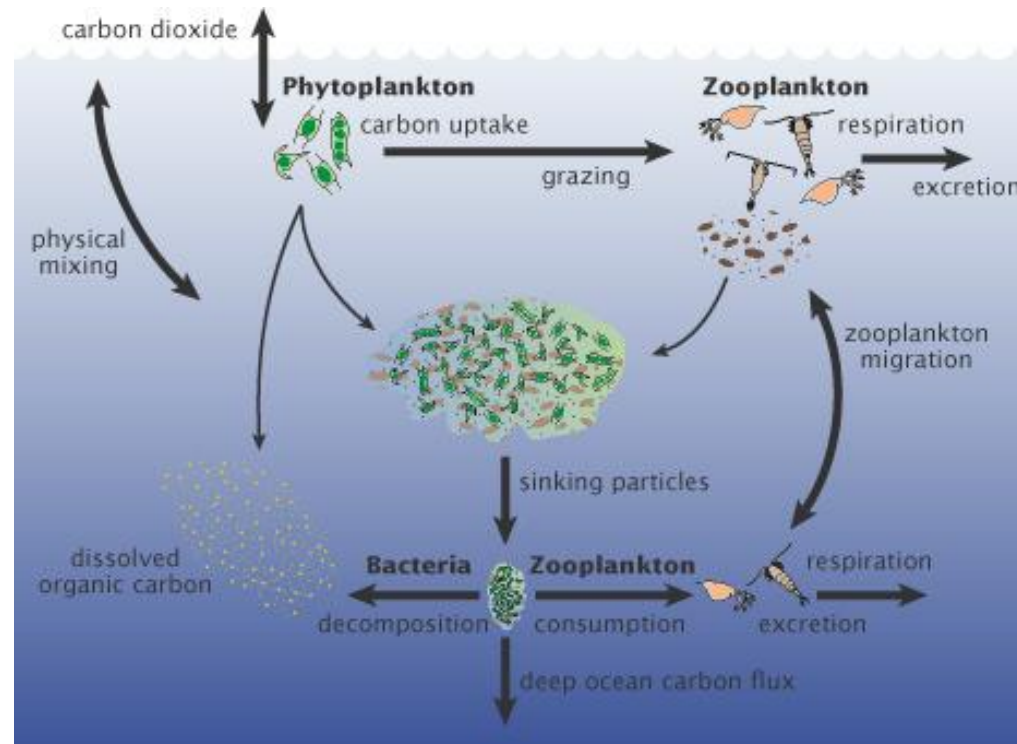


# EFFECTS OF OCEAN ACIDIFICATION AT SPECIES LEVEL

## Calcification

### *Sinking:*

- When calcifying organisms die, their shells sink and contribute to marine sediments
- The  $\text{CaCO}_3$  structures are heavier than the surrounding water and sink to the deeper ocean



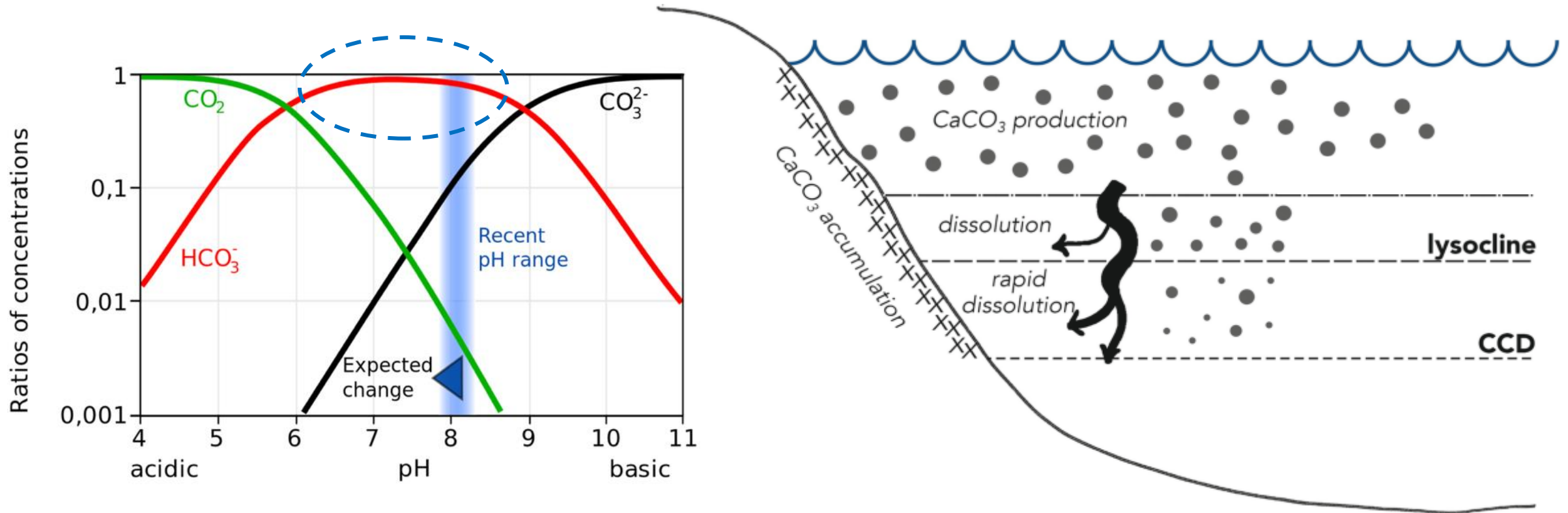


# EFFECTS OF OCEAN ACIDIFICATION AT SPECIES LEVEL

## Calcification

### *Lysocline:*

- The sinking  $\text{CaCO}_3$  structures dissolve at greater depths
- The depth where the rate of dissolution of calcium carbonate (specifically calcite) increases ~ Lysocline
- Transition zone where calcite becomes unstable and starts to dissolve more rapidly
- This increases  $[\text{HCO}_3^-]$ :  $\text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2 \leftrightarrow \text{Ca}^{2+} + 2\text{HCO}_3^- \sim \text{increases pH} \rightarrow \text{Maintains pH levels}$

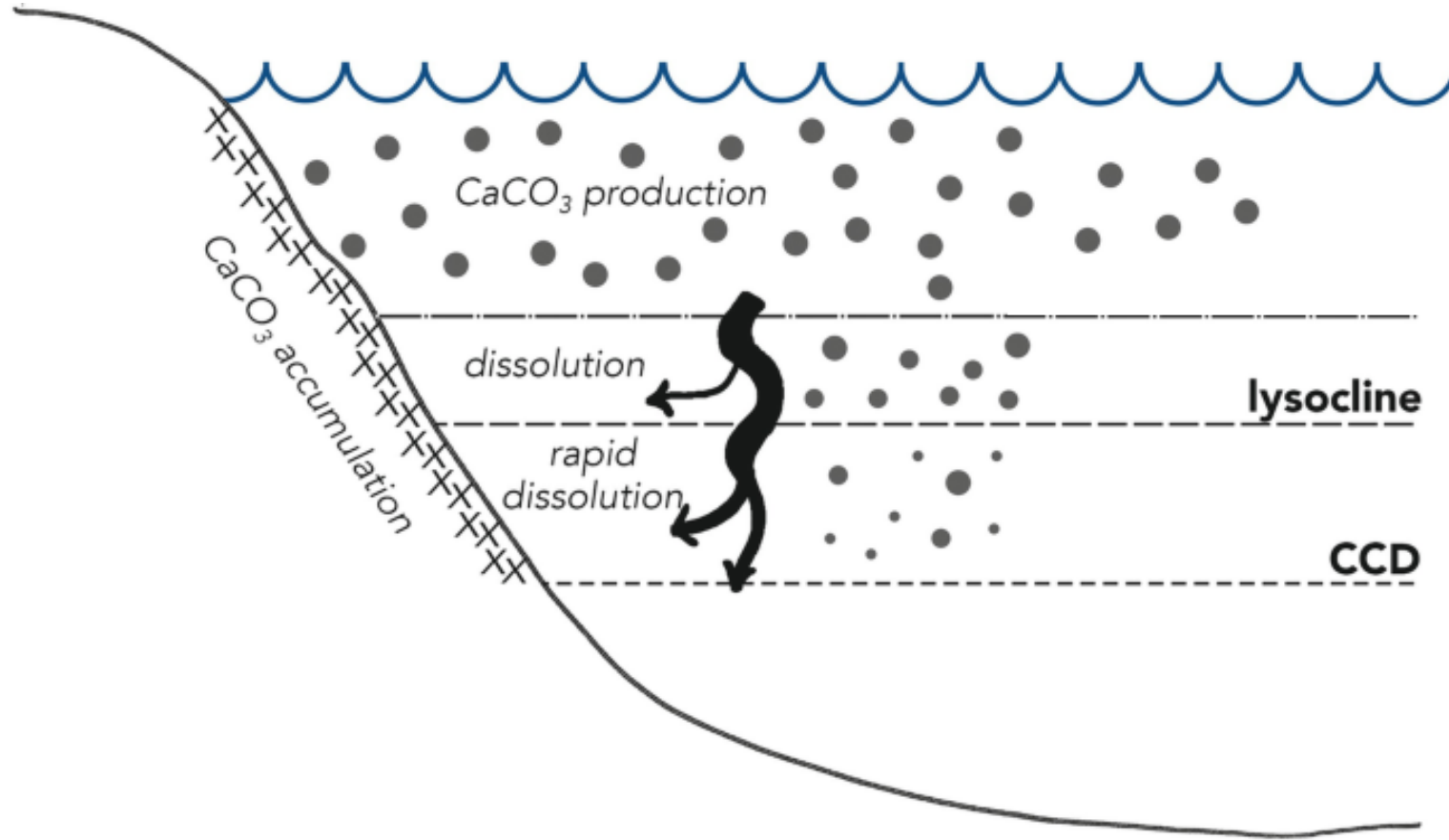


# EFFECTS OF OCEAN ACIDIFICATION AT SPECIES LEVEL

## Calcification

### *Carbonate Compensation Depth (CCD):*

- At a certain depth, the rate at which calcium carbonate dissolves becomes equal to the rate at which it is being supplied from above ~ results in net accumulation of sediments
- The CCD represents the depth below which little or no calcium carbonate accumulates



# EFFECTS OF OCEAN ACIDIFICATION AT SPECIES LEVEL

## Calcification

- Impacts on calcifying species:
  - Increased shell dissolution
  - Thinner, malformed, and/or weaker shells
  - Reduced growth rates
  - Increased vulnerability to predation and environmental stress



# EFFECTS OF OCEAN ACIDIFICATION AT SPECIES LEVEL

## Acid-Base Regulation

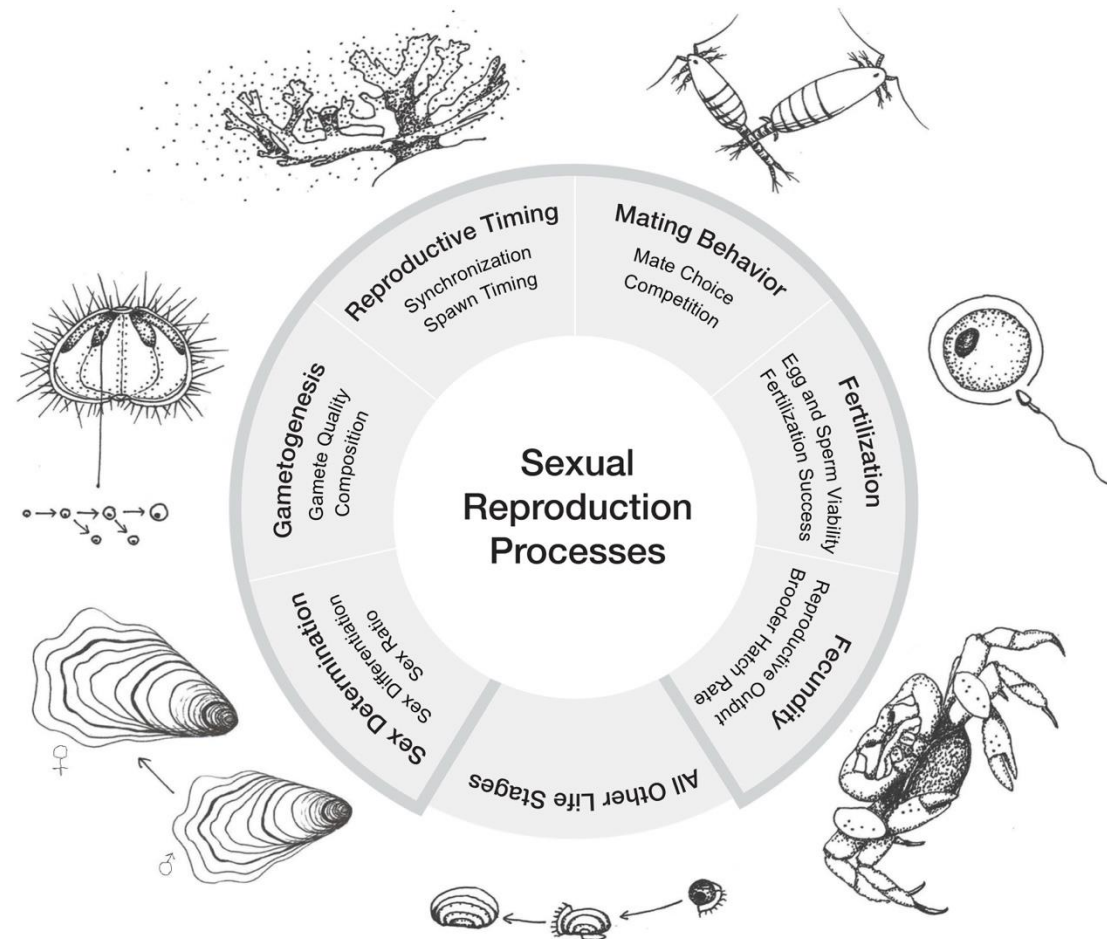
- Ocean acidification affects acid-base regulation in marine organisms
  - Ability of marine organisms to regulate their internal pH
- ***Increased Energy Expenditure:*** Marine organisms expend energy to maintain their internal pH balance, and this effort can be exacerbated by ocean acidification
- ***Trade-offs in Body Systems:*** The need to regulate acid-base balance can come at the expense of other physiological processes, such as growth, reproduction, and energy metabolism
- ***Reduced Performance:*** The capacity for acid-base regulation is crucial for overall animal performance, and disruptions can lead to reduced growth, reproduction, and energy metabolism



# EFFECTS OF OCEAN ACIDIFICATION AT SPECIES LEVEL

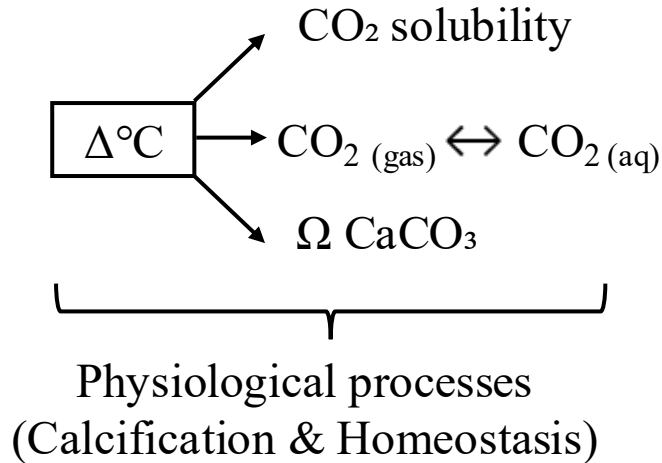
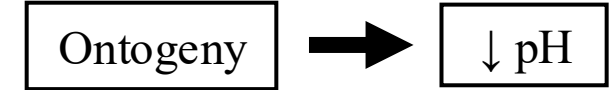
## Development and Reproduction

- Many marine larvae are more sensitive to low pH than adults
- Reduced fertilization success, delayed development, and lower survival rates observed in several species



# EFFECTS OF OCEAN ACIDIFICATION AT SPECIES LEVEL

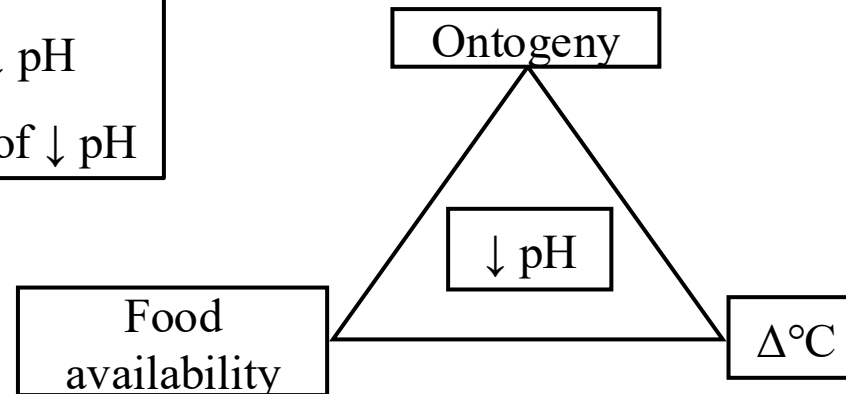
## Tolerance to acidification



- Physiological processes are energy demanding
- Tolerance to  $\downarrow \text{pH}$  ~ determined by energetic context
- Which is modulated by food availability
- $\uparrow$  food supply  $\rightarrow$  moderate effects of  $\downarrow \text{pH}$

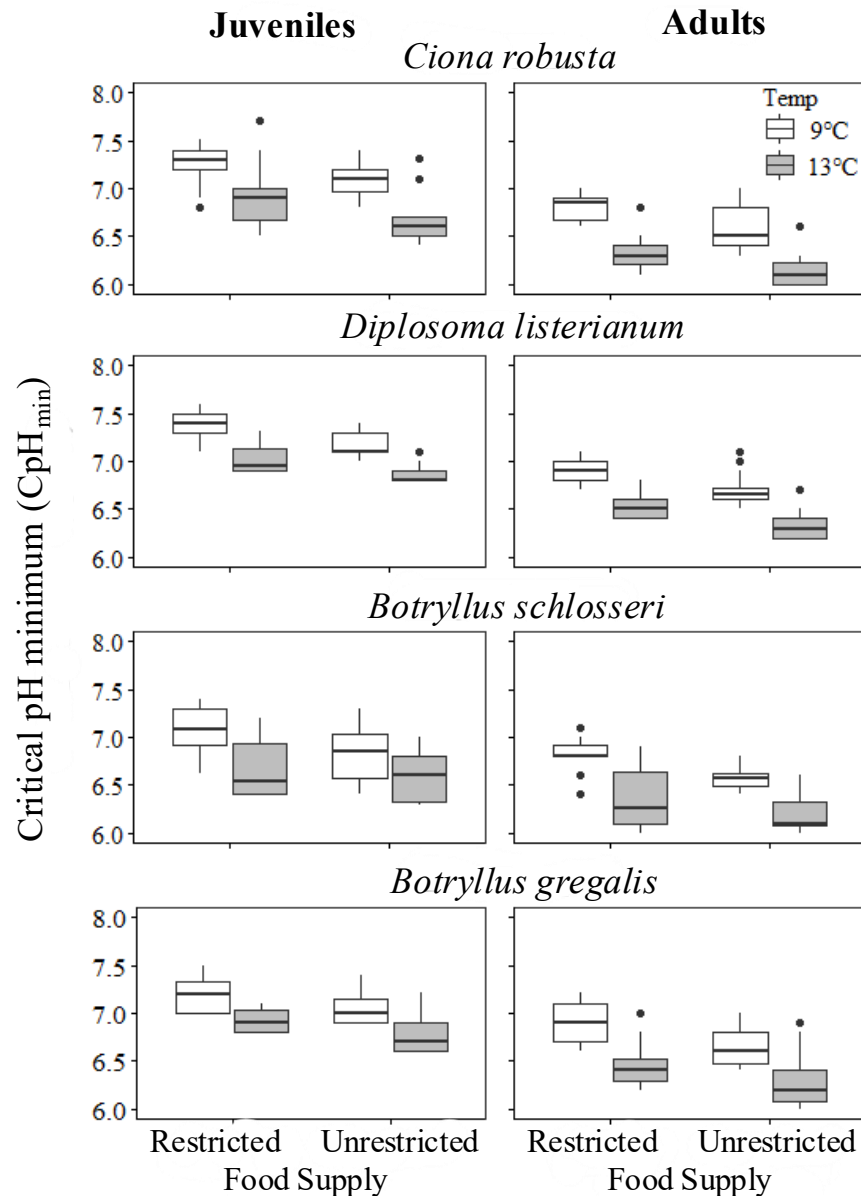
- Tolerance to  $\downarrow \text{pH}$  is life stage-specific
- Larvae & juveniles are more sensitive to  $\downarrow \text{pH}$  than adults

$\uparrow \text{ }^\circ\text{C} \rightarrow \uparrow$  metabolism under food-replete conditions  $\rightarrow$  moderate effects of  $\downarrow \text{pH}$   
 $\uparrow \text{ }^\circ\text{C} > \text{CT}_{\text{max}} \rightarrow$  exacerbate effects of  $\downarrow \text{pH}$



- In nature, these factors do not occur in isolation
- Co-occurring stressors determine tolerance to acidification

# EFFECTS OF OCEAN ACIDIFICATION AT SPECIES LEVEL



MARINE BIOLOGY RESEARCH  
<https://doi.org/10.1080/17451000.2024.2390528>



Taylor & Francis  
Taylor & Francis Group

ORIGINAL ARTICLE

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## Altered temperature and restricted food supply reduce pH tolerance in both young and adult ascidians

Phikolomzi Matikinca <sup>a,b</sup> and Tamara B. Robinson <sup>a</sup>

### *Life stage*

- Juveniles are less tolerant to acidification than adults

### *Temperature or food supply*

- Cooling **OR** restricted food supply reduce tolerance to acidification

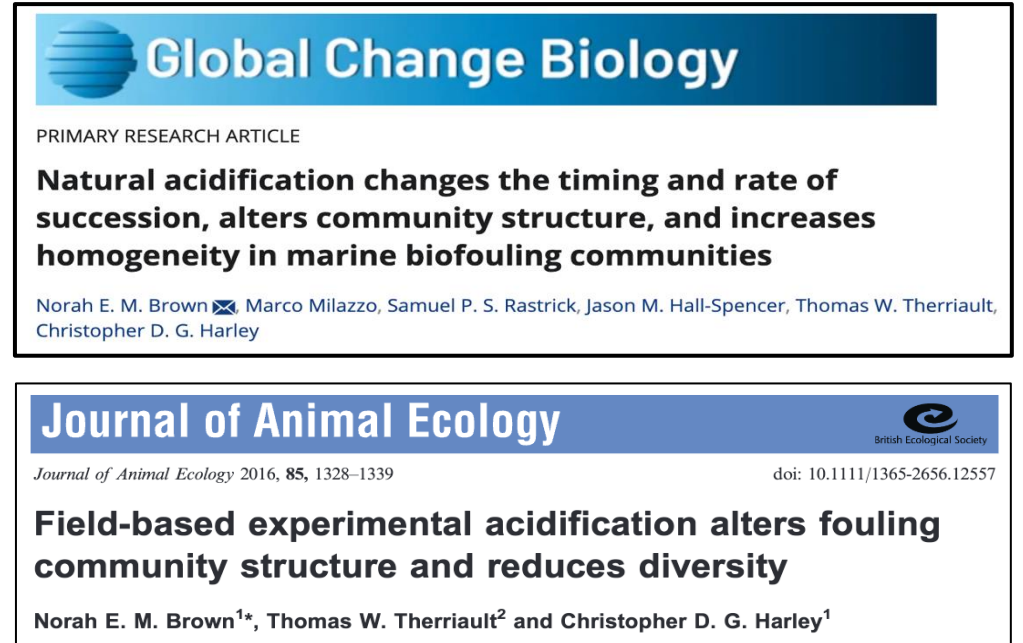
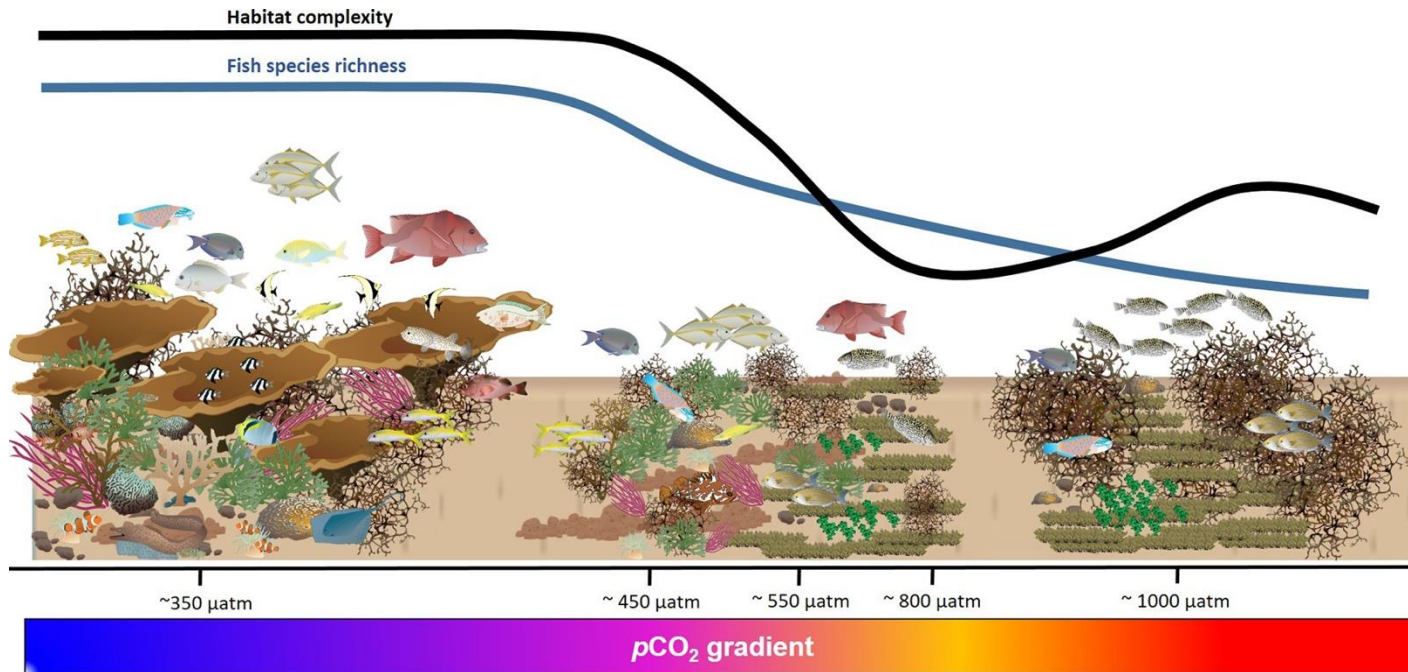
### *Additive effect – temperature and food supply*

For all species and both life stages:

- Cooling **&** restricted food supply reduce to tolerance

# EFFECTS OF OCEAN ACIDIFICATION AT COMMUNITY LEVEL

## Shifts in Species Composition



- Alters recruitment and succession
- Reduces species diversity and abundance ~ changes in overall community structure
- Could result in more simplified communities, dominated by species with broad tolerance ranges
- Calcifiers might be severely affected compared to non-calcifying biota
- Which could facilitate competition, where some species are outcompeted by others



# EFFECTS OF OCEAN ACIDIFICATION AT COMMUNITY LEVEL

Marine Biodiversity (2024) 54:27  
<https://doi.org/10.1007/s12526-024-01420-0>

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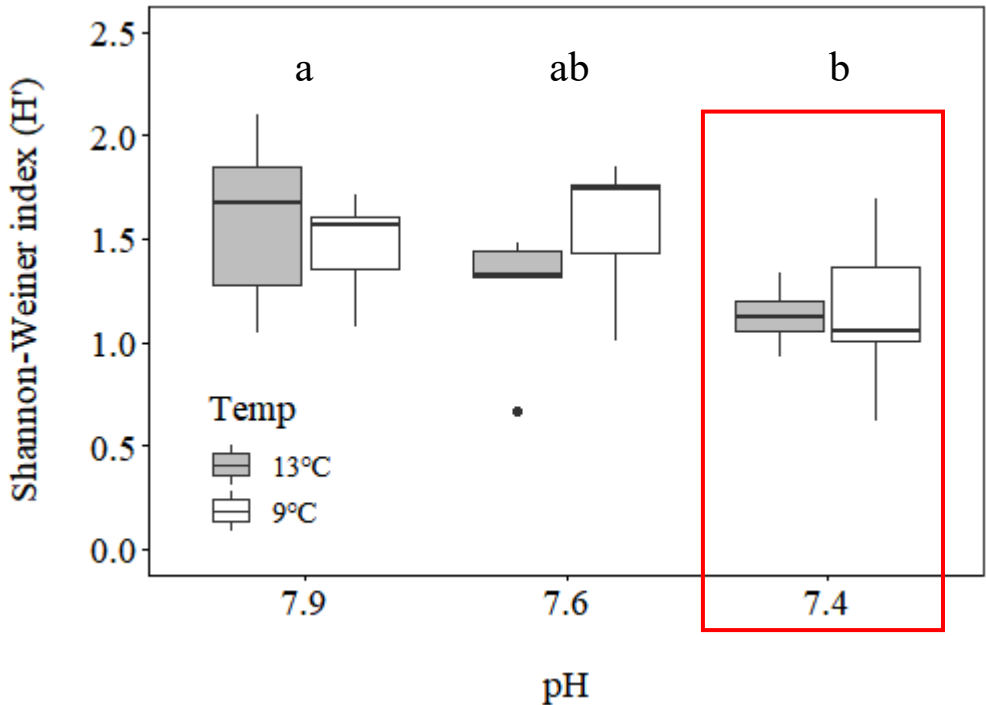
ORIGINAL PAPER



**Fouling communities from the South African west coast are vulnerable to cooling and ocean acidification**

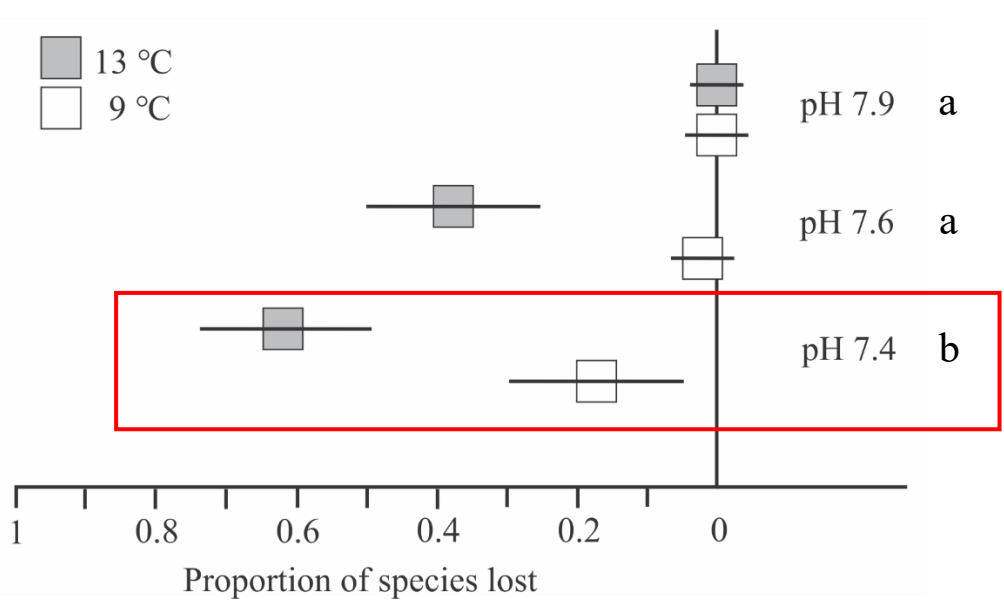
Phikolomzi Matikinca<sup>1,2</sup> · Tamara B. Robinson<sup>1</sup>

| Factor           | <i>p</i> -value |
|------------------|-----------------|
| pH               | < 0.05          |
| Temperature      | ns              |
| Temperature × pH | ns              |



| Factor           | <i>p</i> -value |
|------------------|-----------------|
| pH               | 0.001           |
| Temperature      | ns              |
| Temperature × pH | ns              |

**Mortality**





**THANK YOU**